# ASTUTE RAG: Overcoming Imperfect Retrieval Augmentation and Knowledge Conflicts for Large Language Models

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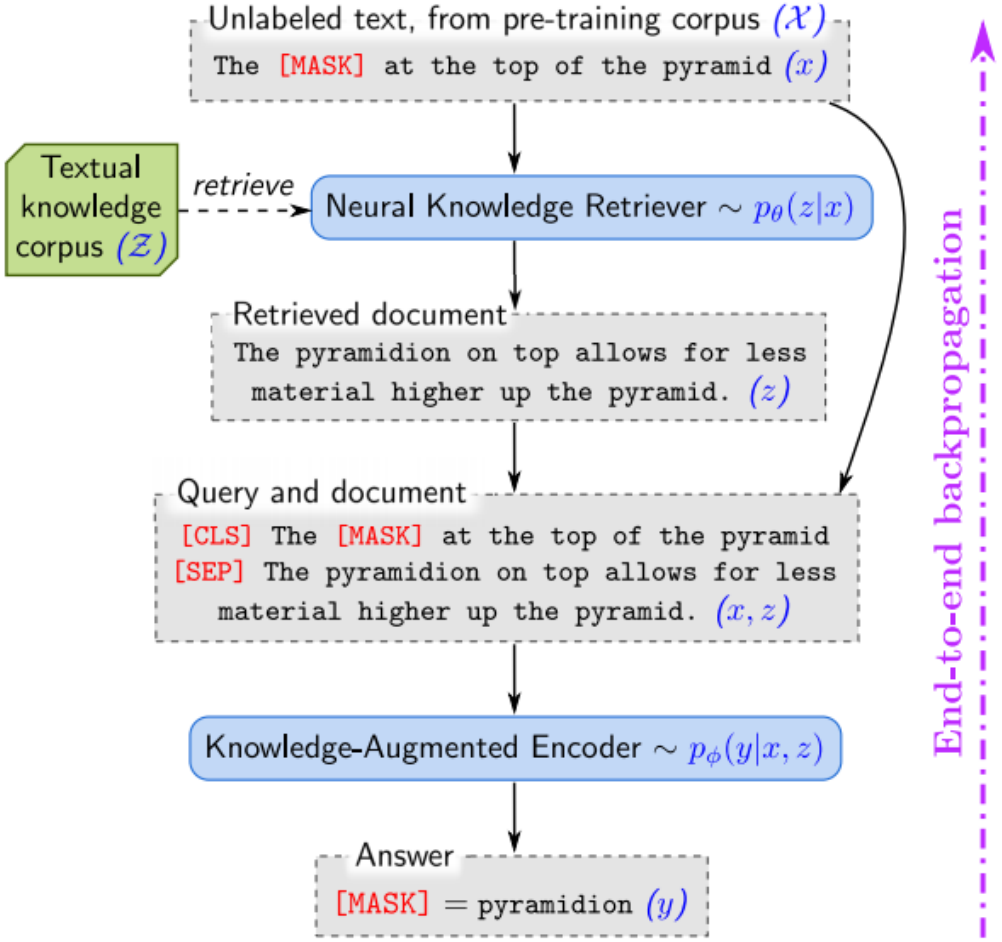
# Background

- **Retrieval-augmented generation (RAG)**




- RAG is commonly used for large language models (LLMs) to tackle **knowledge-intensive tasks** (Guu et al., 2020)

- **Knowledge-intensive Tasks (Prior works)**
  - Mainly leverage RAG to address the inherent knowledge limitations of LLMs
  - Effectively integrate missing information and grounding to reliable sources

**REALM (Guu et al., 2020)**

- **Contribution**
  - Model the Retrieve-then-predict approach as a latent variable language model and optimizing the marginal likelihood

**Answer Likelihood**

$$p(y \mid x) = \sum_{z \in \mathcal{Z}} p(y \mid z, x)\, p(z \mid x)$$

Kelvin Guu et al. Retrieval augmented language model pre-training. ICML 2020.

# Background

- **The drawback of RAG: Imperfect retrieval**
  - Rely on imperfect retrieval, including irrelevant, misleading, or even malicious information, which eventually leads to inaccurate LLM responses

**Model: Claude 3.5 Sonnet(Year 2024, Size 175B)**

*Consistent and accurate answers from both sides. No improvement space.*

| | RAG correct | RAG incorrect |
|---|---|---|
| LLM correct | 45.4% | 9.1% |
| LLM incorrect | 10.1% | 35.4% |

**Previous work leverages RAG to address LLMs' knowledge gap.**

*Zonia receives from Reuben a letter in the play.*

*Zonia receives from Reuben a kiss in the play.*

**Literature: Action from American Play**

**It is also crucial to mitigate RAG errors with LLMs' internal knowledge.**

*China's One Child Policy ended on January 1, 2016.*

*China announced the end of its one-child policy in late 2015.*

**News: Historical Events**

*LLM* *RAG*

*Blue: Correct*

*Red: Incorrect or misleading*

***Both sides are wrong. It is hard to improve, but combining internal and external knowledge may help.***

# Background

- **The drawback of RAG: Knowledge Conflicts**
  - Retrieval augmentation can confuse LLMs when retrieved passages are conflicting with LLMs' parametric knowledge (Xie et al., 2024)
  - **Parametric memory**
    - LLMs have formed a wealth of commonsense and factual knowledge
    - Such memory may be inaccurate or become outdated
    - Cause: misinformation in the pre-training corpus or the static nature of the memory
    - Major cause: Hallucinations
  - **Counter-memory**
    - External evidence that conflicts with parametric memory
  - **External evidence**
    - Tool or retrieval augmentation
    - Provide external information as new evidence, such as ChatGPT Plugins and New Bing

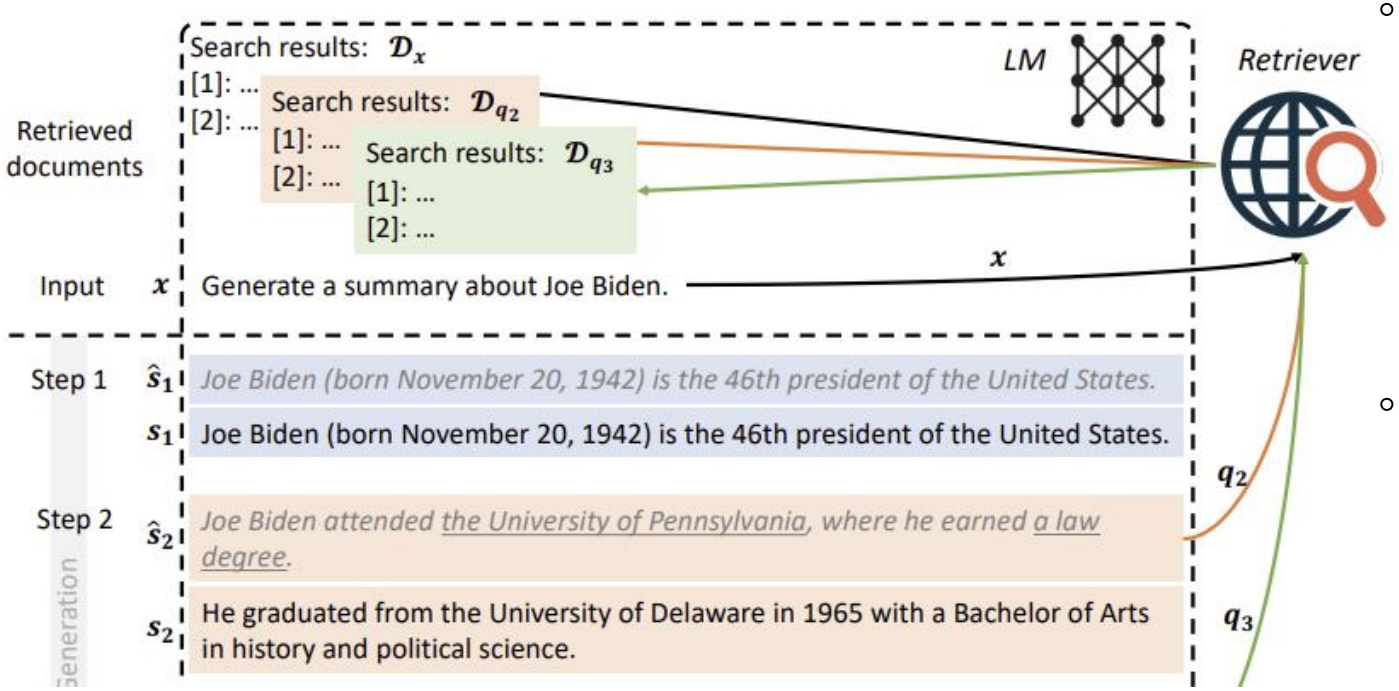


. Jian Xie et al. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts. ICLR. 2024.

# Background

## • Challenge for Imperfect retrieval

**FLARE (Jiang et al., 2023)**




- The retrieval approaches, such as dynamic and iterative retrieval (Jiang et al., 2023)
  - Corpus quality limitations (Shao et al., 2024)
  - The reliability of retrievers (Dai et al., 2024)
  - The complexity of queries (Su et al., 2024)

- The generation stage
  - Seek to reduce the negative impact of noisy retrieved passages (Xiang et al., 2024; Wei et al., 2024)

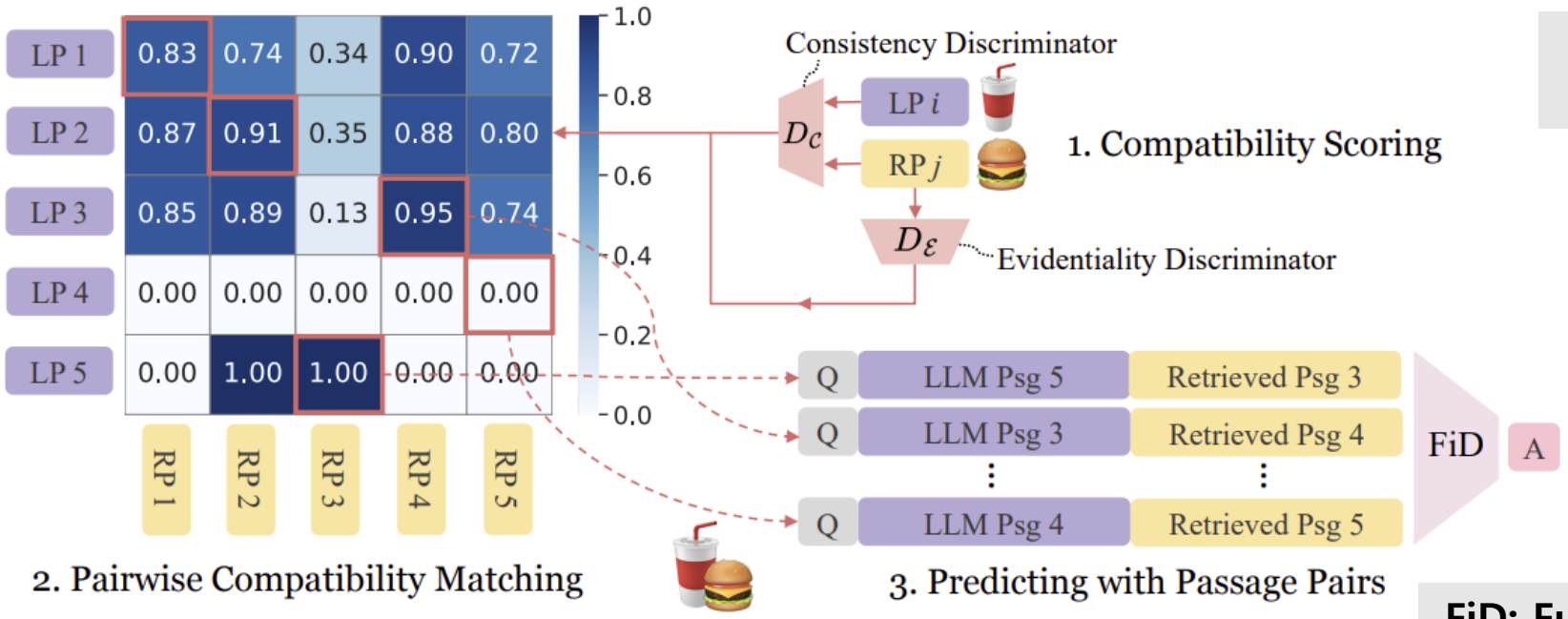


Zhengbao Jiang et al. Active Retrieval Augmented Generation. EMNLP. 2023.

# Background

- **Challenge for Knowledge Conflicts**
  - Complement retrieved passages with LLM-generated passages (Zhang et al., 2023)
  - Deactivate RAG when the retrieved passages are of insufficient quality (Xu et al., 2024; Mallen et al., 2023; Jeong et al., 2024)

**COMBO (Zhang et al., 2023)**




**Compatibility Score = Evidentiality * Consistency**

**FiD: Fusion-in-Decoder (Izacard et al., 2021)**

Yunxiang Zhang et al. Merging Generated and Retrieved Knowledge for Open-Domain QA. EMNLP. 2023.
Gautier Izacard et al. Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering. EACL. 2021.

# Background

- **Quantitative analyses lack on crucial real-world aspects**
  - (i) The relation between retrieval quality and occurrence of knowledge conflicts
  - (ii) The extent to which retrieved passages and LLMs' parametric knowledge can correct each other
  - Existing approaches have not yet yielded **a training-free method**
  - Training-free method
    - Explicitly analyze conflicting knowledge across various internal and external sources
    - Achieve worst-case robustness for black-box LLMs

# Introduction

- **Our findings**
  - Comprehensive analyses to investigate the relation between imperfect retrieval and knowledge conflicts
  - Examine the frequency of external and LLMs' internal knowledge mutually correcting each other
  - Observe that imperfect retrieval is widespread even with an adept real-world search engine, leading to the impeded performance of RAG
  - Observation on a diverse range of general, domain-specific, and long-tail questions
    - **NQ** (Kwiatkowski et al., 2019), **TriviaQA** (Joshi et al., 2017), **BioASQ** (Tsatsaronis et al., 2015), and **PopQA** (Mallen et al., 2023)

Tom Kwiatkowski et al. Natural questions: a benchmark for question answering research. TACL. 2019.
Mandar Joshi et al. Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension. ACL. 2017.
George Tsatsaronis et al. An overview of the bioasq large-scale biomedical semantic indexing and question answering competition. BMC bioinformatics, 16:1–28. 2015.
Alex Mallen et al. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. ACL. 2023.

# Introduction

- **Our findings**
  - Mutual correction between the LLM's knowledge and external knowledge
  - ➡ It is crucial Recovering from RAG failures
  - Emphasize the potential severity of imperfect retrieval in real-world RAG
  - Highlight the widespread existence of knowledge conflicts as the bottleneck

# Introduction

- **ASTUTE RAG**
  - RAG approach designed for resilience to imperfect retrieval augmentation,
  - while preserving RAG grounding effect when retrieval is reliable
  - Effectively differentiates between consistent and conflicting information from the LLM's internal knowledge and the externally retrieved passages
  - Assess their reliability, and ensures proper integration of trustworthy information

**astute [ əˈstuːt ]**

**It was an astute move to sell the shares then.**

# Introduction

- **ASTUTE RAG**
  - Proposes answers based on consistent information and compares them to determine the final answer
    - Combine consistent information
    - Identify conflicting information
    - Filter out irrelevant information
  - Our experiments
    - LLMs: Claude 3.5 Sonnet, Gemini 1.5 Pro, MistralLarge (128B), MistralNemo (12B)
    - Consistently outperforms baselines across different retrieval quality levels
    - Achieve performance comparable retrieval-free mode of LLMs under the worst-case scenario where all retrieved passages are unhelpful

# The Pitfall of RAG

- **Our analysis**
  - (1) Retrieval quality
  - (2) The occurrence of knowledge conflicts, their relationship
  - (3) The mutual correction between external and internal knowledge

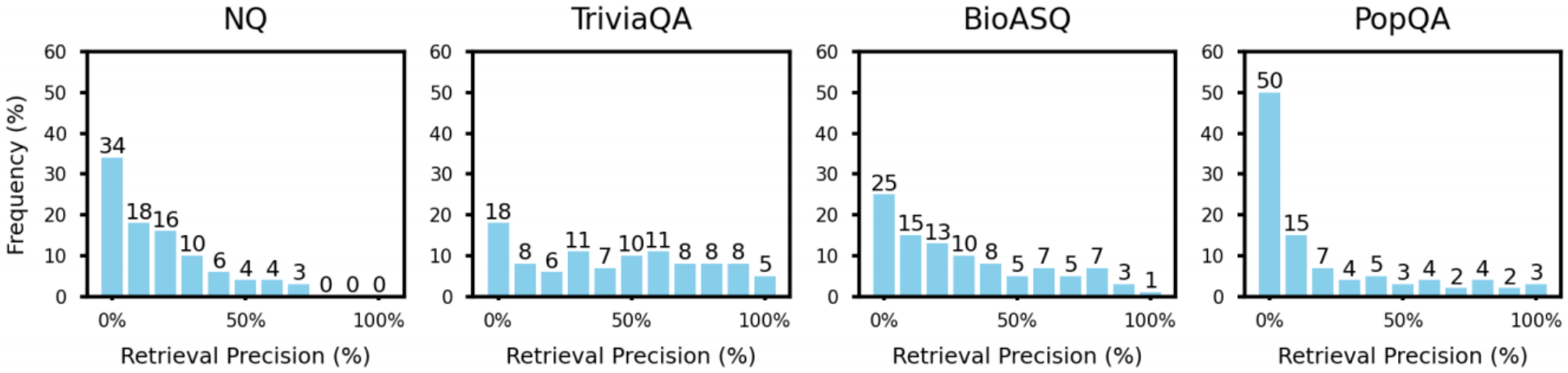
  - It is based on real-world retrieval results
    - Google Search2 as the retriever
    - The Web as the corpus

  - Overall, we sample 1K instances, each with 10 retrieved passages

# The Pitfall of RAG

- **Imperfect retrieval and knowledge conflicts are common and harmful**
  - The retrieval precision is generally low - roughly 70% retrieved passages do not directly contain true answers

➡ Imperfect retrieval (samples with low retrieval precision) is prevalent in real-world RAG



# The Pitfall of RAG

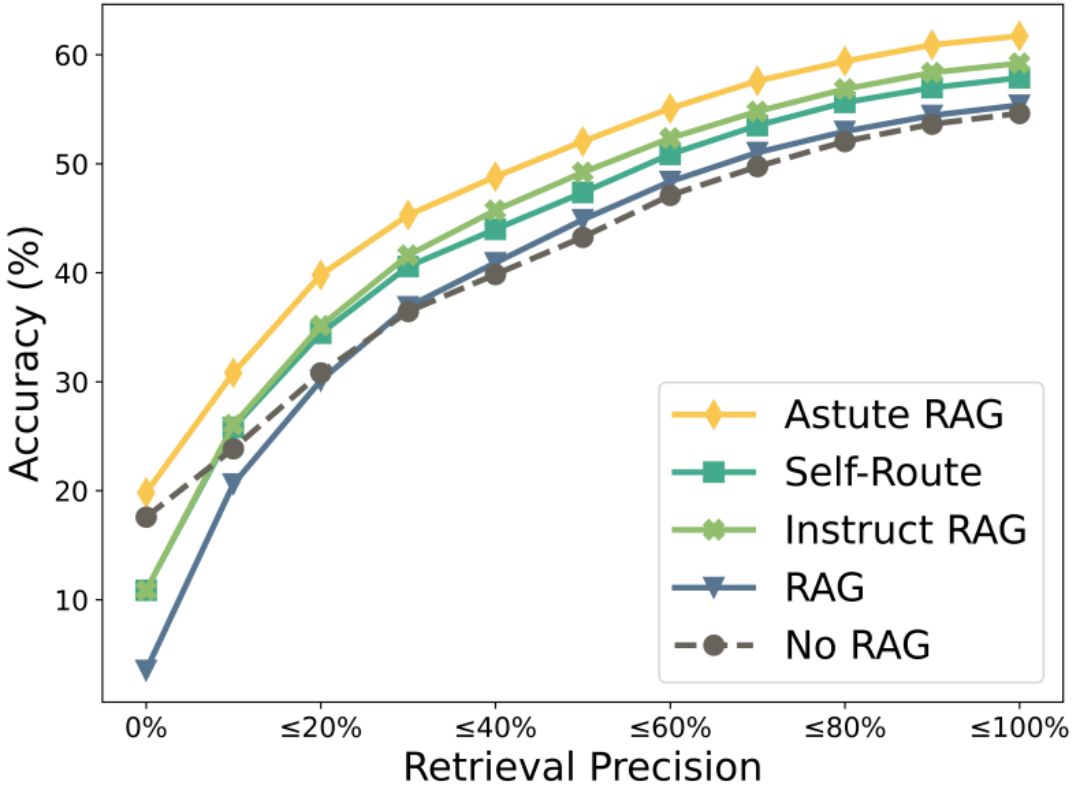
- **Imperfect retrieval and knowledge conflicts are common and harmful**
  - With Claude 3.5 Sonnet as the LLM shows that 19.2% of the overall data exhibit knowledge conflicts

| | RAG correct | RAG incorrect |
|---|---|---|
| LLM correct | 45.4% | 9.1% |
| LLM incorrect | 10.1% | 35.4% |

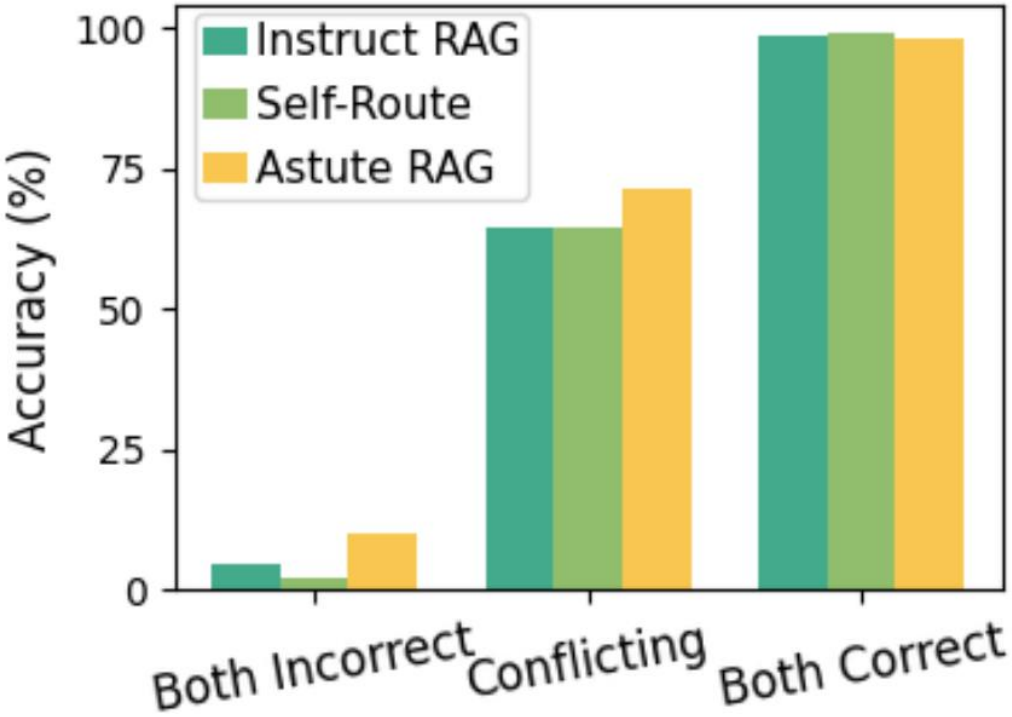
# The Pitfall of RAG

- **Imperfect retrieval and knowledge conflicts are common and harmful**

**Performance across different retrieval precision buckets**

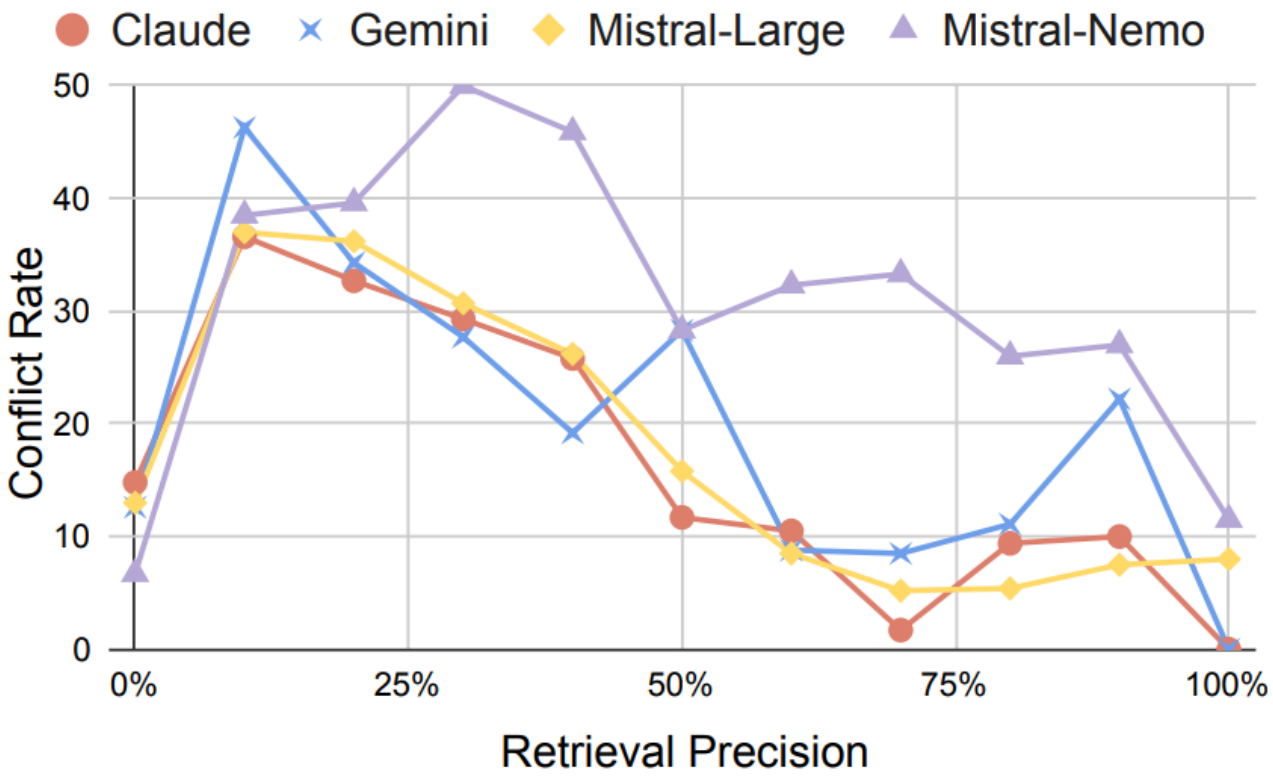



**Performance on conflicting and consistent instances between No RAG and RAG**



# The Pitfall of RAG

- **Lower retrieval precision increases knowledge conflicts in general**




- **A notable feature**
- When retrieval precision is 0%, conflict rates tend to be significantly lower

➡ Limited external knowledge for the query results in more irrelevant passages rather than incorrect ones

# The Pitfall of RAG

- **Internal and external knowledge can correct each other to a comparable extent**

| | RAG correct | RAG incorrect |
|---|---|---|
| LLM correct | 45.4% | 9.1% |
| LLM incorrect | 10.1% | 35.4% |

- **47.4%** $= \dfrac{\text{P(RAG Incorrect \& LLM Correct)}}{\text{P(RAG Incorrect \& LLM Correct)} + \text{P(RAG Correct \& LLM Incorrect)}}$
- The internal knowledge is correct on 47.4% of conflicting cases
- The external knowledge is correct on the remaining 52.6%

➡

- Emphasize the importance of effectively combining the internal and external knowledge

➡

- Overcome the inherent limitation of relying solely on either source

# ASTUTE RAG: Overview

- **Method**
  - **1. Overview of ASTUTE RAG**

  - **2. Steps of ASTUTE RAG**
  - 2.1. Adaptive generation of internal knowledge
  - 2.2. Source-aware knowledge consolidation
  - 2.3. Answer finalization

# ASTUTE RAG: Overview

- **ASTUTE RAG framework**

**Algorithm 1** ASTUTE RAG

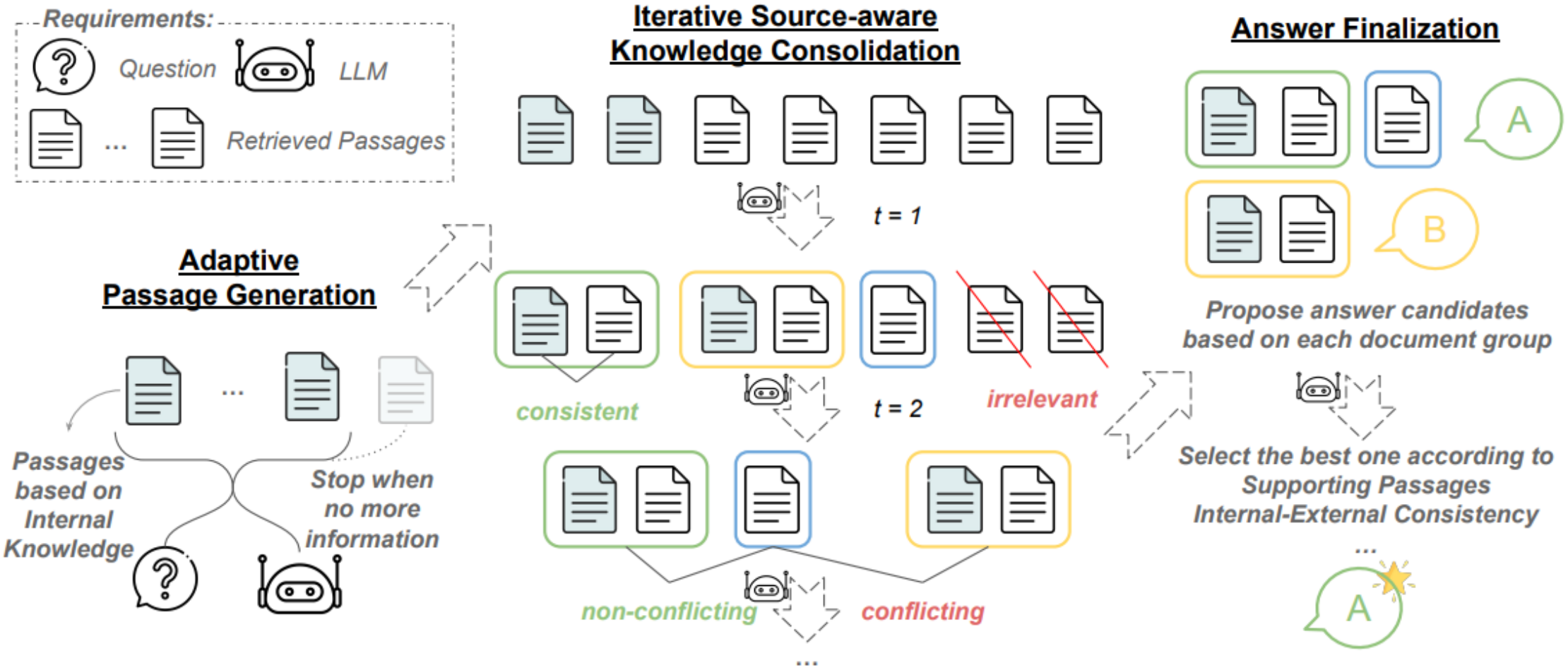
**Require:** Query $q$, Retrieved Passages $E = [e_1, \ldots, e_n]$, Large Language Model $\mathcal{M}$, Number of Iteration $t$, Max Number of Generated Passages $\hat{m}$, Prompt Templates $p_{gen}, p_{con}, p_{ans}$

Adaptively generate passages: $I \leftarrow \mathcal{M}(p_{gen}, q, \hat{m})$ ▷ Sec. 4.2
Combine internal and external passages: $D_0 \leftarrow E \oplus I$
Assign passage sources: $S_0 \leftarrow [\mathbb{1}_{\{d \in E\}} \text{for } d \text{ in } D_0]$
**if** $t > 1$ **then**
  **for** $j = 1, \ldots, t-1$ **do** ▷ Sec. 4.3
    Consolidate knowledge: $\langle D_{j+1}, S_{j+1} \rangle \leftarrow \mathcal{M}(p_{con}, q, \langle D_0, S_0 \rangle, \langle D_j, S_j \rangle)$
  **end for**
  Finally consolidate and answer: $a \leftarrow \mathcal{M}(p_{ans}, q, \langle D_0, S_0 \rangle, \langle D_{t-1}, S_{t-1} \rangle)$ ▷ Sec. 4.4
**else**
  Consolidate knowledge and finalize the answer: $a \leftarrow \mathcal{M}(p_{ans}, q, \langle D_0, S_0 \rangle)$
**end if**
**return** $a$

- A pre-trained LLM $M$
  - Accessible through prediction-only APIs, encompassing commercial black-box ones

# ASTUTE RAG: Overview

- **ASTUTE RAG framework**

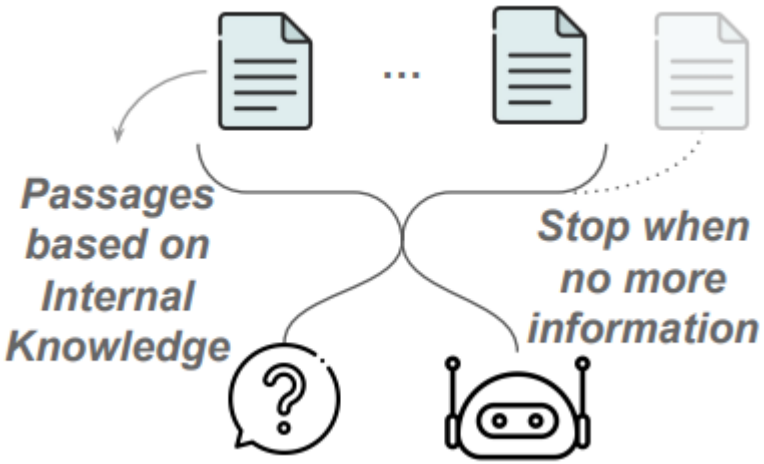


# Adaptive Generation of Internal Knowledge

- **Prompt $p_{gen}$**
  - Inspiration: (Yu et al., 2023a) generate passages by prompting

Generate a document that provides accurate and relevant information to answer the given question. If the information is unclear or uncertain, explicitly state 'I don't know' to avoid any hallucinations.

Question: {question} Document:

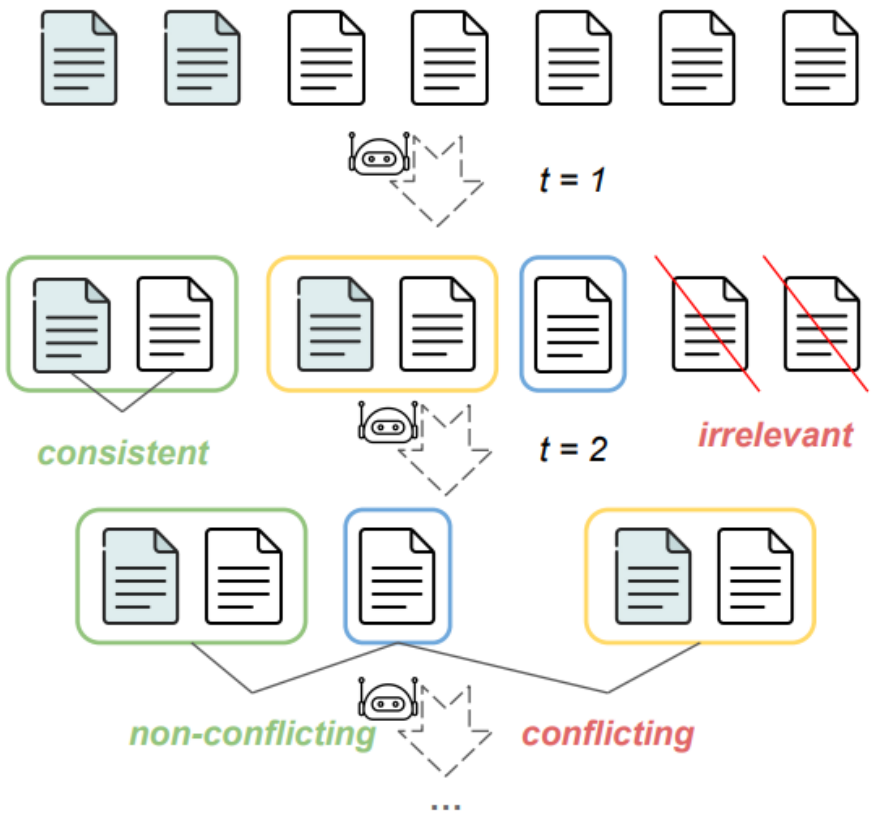



- Constitutional principles
  - Accurate, relevant, hallucination-free
  - Inspiration: Constitutional AI (Bai et al., 2022)
- Adaptive generation $I = [i_1, \ldots i_m] = \mathcal{M}(p_{gen}, q, \hat{m})$
  - Rather generating a fix number of passages, we request the LLM to generate at most $m^{\wedge}$ passages

Wenhao Yu et al. Generate rather than retrieve: Large language models are strong context generators. ICLR. 2023.
Yuntao Bai et al. Constitutional ai: Harmlessness from ai feedback. arXiv preprint arXiv:2212.08073. 2022.

# Iterative Source-aware Knowledge Consolidation

- **Prompt $p_{con}$**




Task: Consolidate information from both your own memorized documents and externally retrieved documents in response to the given question.

* For documents that provide consistent information, cluster them together and summarize the key details into a single, concise document.
* For documents with conflicting information, separate them into distinct documents, ensuring each captures the unique perspective or data.
* Exclude any information irrelevant to the query.

For each new document created, clearly indicate:
* Whether the source was from memory or an external retrieval.
* The original document numbers for transparency.

Initial Context: {context}
Last Context: {context}
Question: {question}
New Context:

**Combine Knowledge**

$$D_0 = E \oplus I$$

**Source-awareness: Passage source**

$$S_0 = [\mathbb{1}_{\{d \in E\}} \text{for } d \text{ in } D_0]$$

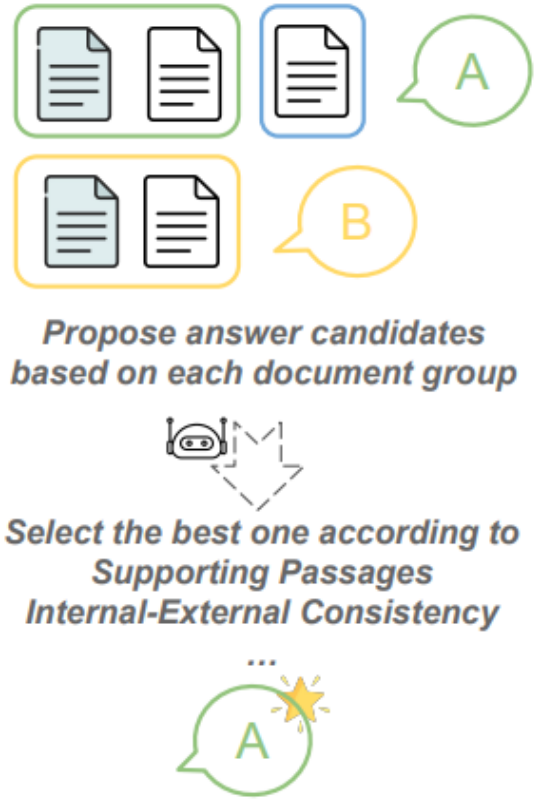
**The regrouped passages**

$$\langle D_{j+1}, S_{j+1} \rangle = \mathcal{M}(p_{con}, q, \langle D_0, S_0 \rangle, \langle D_j, S_j \rangle)$$

Part 7.

# Answer Finalization

- **Prompt $p_{ans}$**




**t = 1** $a = \mathcal{M}(p_{ans}, q, \langle D_0, S_0 \rangle)$

**Last t** $a = \mathcal{M}(p_{ans}, q, \langle D_0, S_0 \rangle, \langle D_t, S_t \rangle)$

Task: Answer a given question using the consolidated information from both your own memorized documents and externally retrieved documents.

Step 1: Consolidate information
* For documents that provide consistent information, cluster them together and summarize the key details into a single, concise document.
* For documents with conflicting information, separate them into distinct documents, ensuring each captures the unique perspective or data.
* Exclude any information irrelevant to the query.
For each new document created, clearly indicate:
* Whether the source was from memory or an external retrieval.
* The original document numbers for transparency.

Step 2: Propose Answers and Assign Confidence
For each group of documents, propose a possible answer and assign a confidence score based on the credibility and agreement of the information.

Step 3: Select the Final Answer
After evaluating all groups, select the most accurate and well-supported answer.
Highlight your exact answer within <ANSWER> your answer </ANSWER>.

Initial Context: {context_init}
[Consolidated Context: {context}] # optional
Question: {question}
Answer:

# Experiments

- **Datasets**
  - General questions, domain-specific questions, long-tail questions, shortform and long-form formats
  - NQ, TriviaQA, BioASQ, PopQA
    - Provide 10 passages collected with Google Search from the Web for each instance
  - Long-form QA: ASQA, RGB
    - Choose the English subset (refined version) focusing on noise robustness
    - For each instance, we select five top negative passages to form a worst-case scenario

# Experiments

- **Datasets with short-form Answers**
  - NQ (Kwiatkowski et al., 2019) and TriviaQA (Joshi et al., 2017)
    - Widely-studied question answering (QA) datasets in general domains
  - BioASQ (Tsatsaronis et al., 2015)
    - Biomedical domain that has demonstrated significant benefits from RAG when general-purpose LLMs are considered
  - PopQA (Mallen et al., 2023)
    - Focus on long-tail knowledge
    - Cover factual information in questions about long-tail entities
    - Use Wikipedia page views as a measure of popularity
    - Challenging datasets for even advanced LLMs to solve without external knowledge

Tom Kwiatkowski et al. Natural questions: a benchmark for question answering research. TACL. 2019.
Mandar Joshi et al. Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension. ACL. 2017.
George Tsatsaronis et al. An overview of the bioasq large-scale biomedical semantic indexing and question answering competition. BMC bioinformatics, 16:1–28. 2015.
Alex Mallen et al. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. ACL. 2023.

# Experiments

- **Datasets with Long-form Answers**
  - ASQA (Stelmakh et al., 2022)
    - Long-form QA dataset for factoid questions
    - Focuses on answering ambiguous factoid questions

**Standard factoid questions (Vladimir Karpukhin et al., 2020)**

Q: **Where** Was The 8th Dalai Lama Born?
A: Lhari Gang in Tibet

**Ambiguous factoid questions (Stelmakh et al., 2022)**

Q: **Who** was the ruler of France in 1830?
A: Charles X was king in the early months of 1830, and Louis-Philippe I became king after the July Revolution.

Ivan Stelmakh et al. Asqa: Factoid questions meet long-form answers. EMNLP. 2022.
Vladimir Karpukhin et al. Dense Passage Retrieval for Open-Domain Question Answering. EMNLP. 2020.

# Experiments

- **Datasets with Long-form Answers**
  - RGB (Chen et al., 2024a)
    - Ensure that the internal knowledge of LLMs does not introduce bias into the evaluation
    - RGB chooses to constructs queries based on the latest news information
    - Then, based on these queries, we use Search API to fetch relevant documents and select most relevant snippets from the content as external retrieved documents

**RGB (Chen et al., 2024a)**

Q: Who was awarded the 2022 Nobel Prize for Physiology and Medicine?
A: Svante Pääbo

Jiawei Chen et al. Benchmarking large language models in retrieval-augmented generation. AAAI. 2024.

# Experiments

- **Models and General Settings**
  - Open Source LLMs
    - Claude 3.5 Sonnet, Gemini 1.5 Pro, MistralLarge (128B), MistralNemo (12B)
  - The generation temperature: 0
  - The maximum output tokens: 1,024
  - All experiments are under the zero-shot setting for controlled evaluation

# Experiments

- **Baselines**
  - USC (Chen et al., 2024b): Reference for using API Calls
    - Self-consistency method that samples multiple LLM responses and aggregates the answers
  - Genread (Yu et al., 2023a)
    - Augment retrieved passages with LLM-generated passages without explicit consolidation process
  - RobustRAG (Xiang et al., 2024)
    - Aggregate answers from independent passages to provide certifiable robustness
    - Use the best-performing keyword aggregation variant
  - InstructRAG (Wei et al., 2024)
    - Instruct the LLM to provide a rationale connecting the answer with information in passages
    - For a fair comparison, no training is applied
  - Self-Route (Xu et al., 2024): Reference for swiching between knowledges
    - Adaptively switches between LLMs with and without RAG

# Experiment

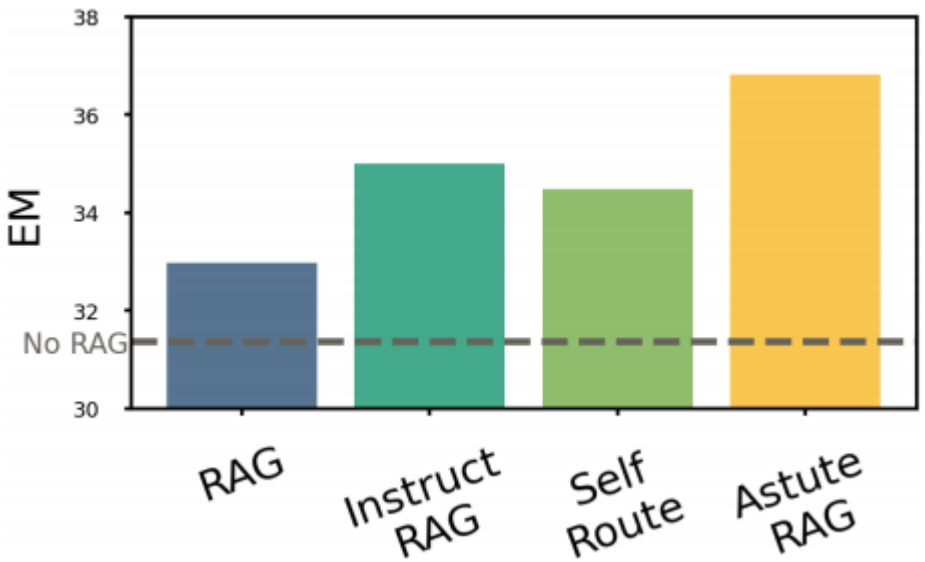
- **Performance under real-world retrieval**

| Method | NQ | TriviaQA | BioASQ | PopQA | Overall | NQ | TriviaQA | BioASQ | PopQA | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| | *Claude 3.5 Sonnet (20240620)* | | | | | *Gemini 1.5 Pro (002)* | | | | |
| No RAG | 47.1 | 82.0 | 50.4 | 29.8 | 54.5 | 44.8 | 80.2 | 45.8 | 25.3 | 51.3 |
| RAG | 44.4 | 76.7 | 58.0 | 36.0 | 55.5 | 42.7 | 76.0 | 55.2 | 33.7 | 53.7 |
| USC (Chen et al., 2024b) | 48.1 | 80.2 | **61.5** | 37.6 | 58.7 | 46.4 | 76.7 | **58.4** | 37.6 | 56.4 |
| GenRead (Yu et al., 2023a) | 42.0 | 74.2 | 57.0 | 34.3 | 53.6 | 45.1 | 77.4 | 54.9 | 34.3 | 54.7 |
| RobustRAG (Xiang et al., 2024) | 47.8 | 78.1 | 56.3 | 37.1 | 56.5 | 34.2 | 67.5 | 44.1 | 32.0 | 45.6[4] |
| InstructRAG (Wei et al., 2024) | 47.1 | 83.0 | 58.0 | 41.0 | 58.8 | 46.8 | 80.6 | 54.9 | 34.8 | 56.1 |
| Self-Route (Xu et al., 2024) | 47.5 | 78.8 | 59.1 | 41.0 | 58.1 | 47.5 | 79.9 | 58.0 | 38.2 | 57.6 |
| ASTUTE RAG | **52.2** | **84.1** | 60.1 | **44.4** | **61.7** | **50.2** | **81.6** | 58.0 | **40.5** | **59.2** |
| | *Mistral-Large (2407), 128B* | | | | | *Mistral-Nemo (2407), 12B* | | | | |
| No RAG | 46.8 | 79.5 | 43.7 | 24.7 | 51.1 | 29.8 | 67.8 | 34.3 | 23.0 | 40.2 |
| RAG | 43.1 | 77.4 | 55.9 | 36.0 | 54.7 | 39.3 | 66.8 | 49.0 | 32.6 | 48.3 |
| USC (Chen et al., 2024b) | **51.2** | 80.9 | **61.5** | 36.0 | 59.5 | 29.5 | 66.1 | 36.0 | 20.2 | 39.6 |
| GenRead (Yu et al., 2023a) | 40.7 | 73.1 | 55.6 | 35.4 | 52.7 | 38.6 | 68.9 | 48.3 | **33.7** | 48.7 |
| RobustRAG (Xiang et al., 2024) | 42.7 | 77.7 | 50.4 | 34.8 | 53.0 | 35.6 | 71.7 | 44.1 | 27.5 | 46.4 |
| InstructRAG (Wei et al., 2024) | 45.4 | 80.6 | 57.3 | 36.5 | 56.7 | 38.3 | 61.8 | 50.4 | 23.6 | 45.5 |
| Self-Route (Xu et al., 2024) | 45.4 | 77.7 | 57.3 | 38.2 | 56.2 | 41.4 | 73.5 | **51.8** | 30.9 | 51.2 |
| ASTUTE RAG | 50.2 | **82.7** | 58.4 | **42.1** | **59.9** | **42.7** | **73.9** | 49.3 | 32.6 | **51.3** |

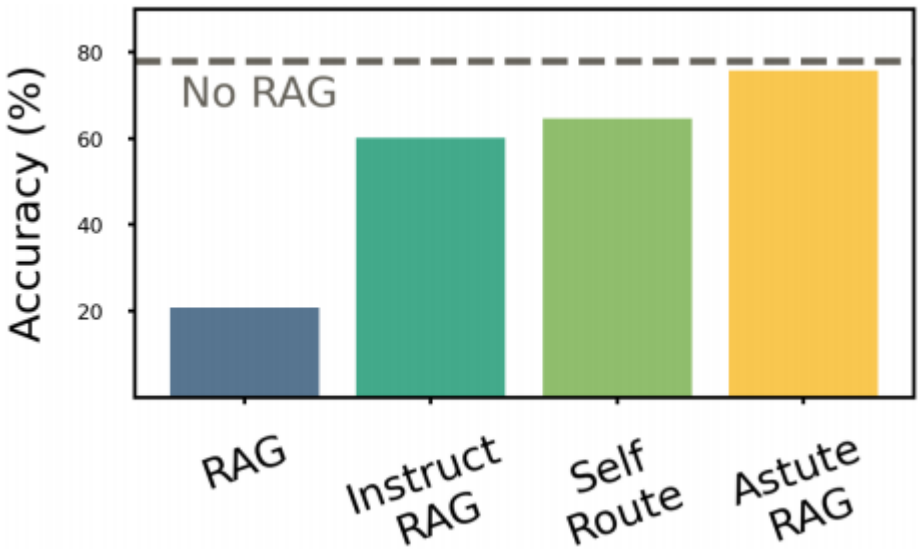
# Experiment

- **Performance on long-form QA**
  - Astute RAG is the only RAG method that reaches a performance close to 'No RAG',

**Performance on ASQA**



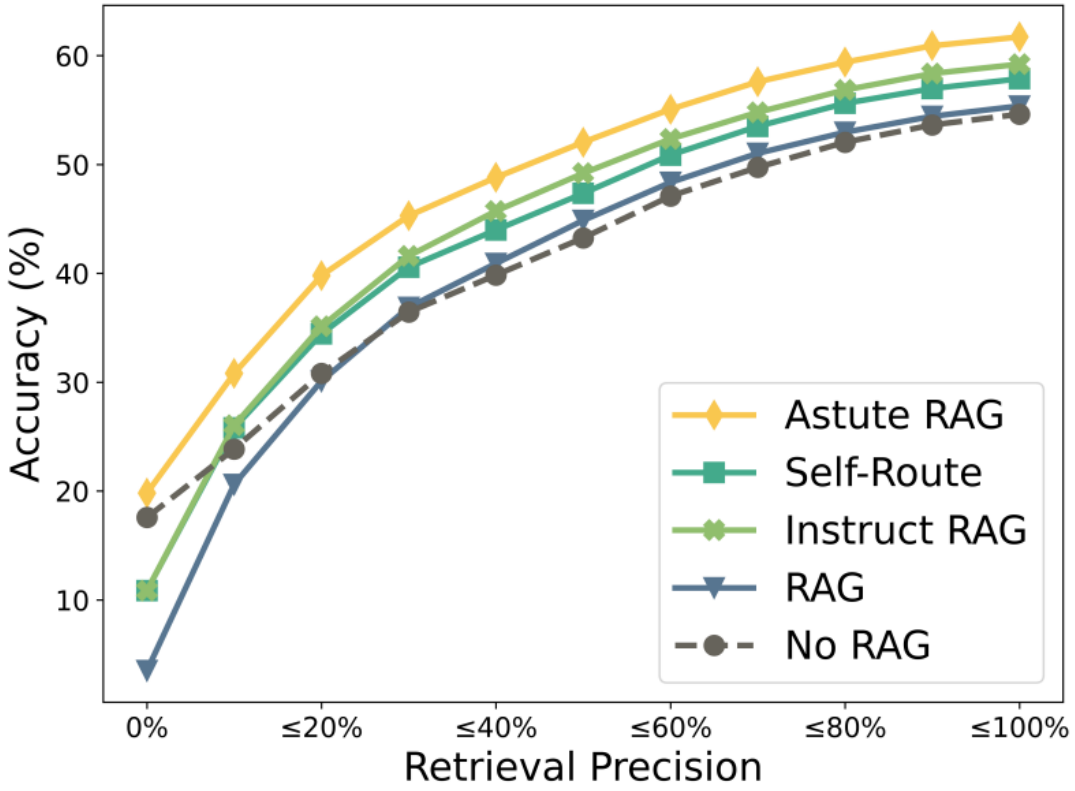

**Worst-case performance of Claude on RGB**



# Analysis

- **The impact of retrieval precision**
  - When the retrieval quality is extremely low (close to zero precision), all other RAG variants underperform the 'No RAG' baseline

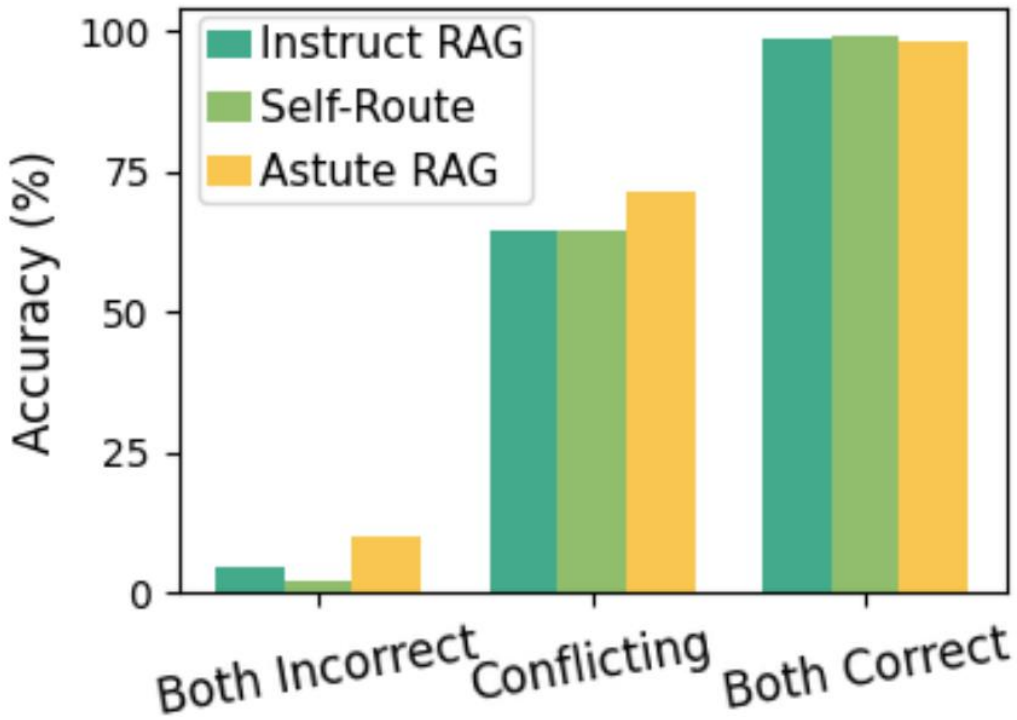
**Performance across different retrieval precision buckets**



# Analysis

- **Addressing knowledge conflicts**
  - Bring performance improvement on the subset where neither internal nor external knowledge alone leads to the correct answer

**Performance on conflicting and consistent instances between No RAG and RAG**



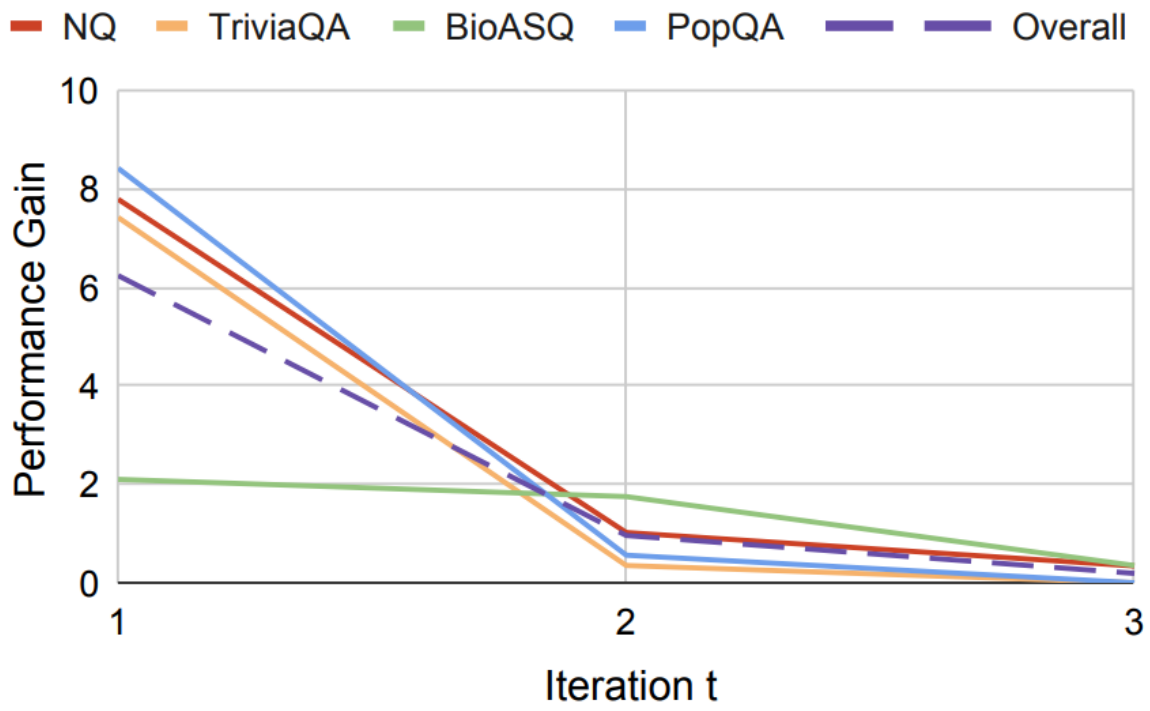

- Split our collected data into three subset according to the answers with and without RAG

- The answers from two can be
- (i) both correct
- (ii) both incorrect
- (iii) conflicting with one being correct

# Analysis

- **Benefits of more consolidation iteration**
  - The magnitude of this improvement diminishes as t increase
  - It indicates that the knowledge has been better presented and less improvement space left after each iteration

**Accuracy improvement when increasing t**

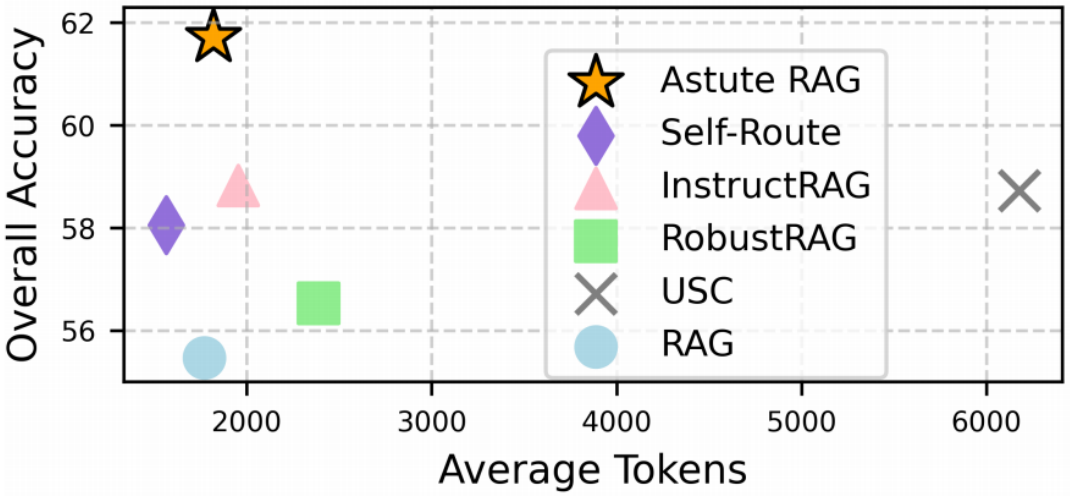


# Analysis

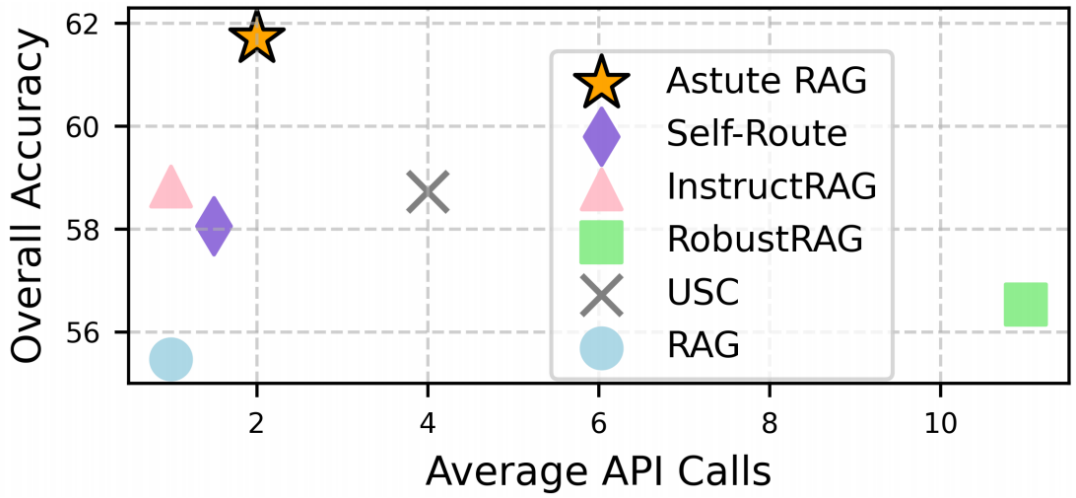
- **Efficiency in tokens consumed and API**
  - As a proxy to overall prediction cost and latency, we present the average number of tokens and API calls used per instance
  - ASTUTE RAG incurs only a marginal cost increase, <5%,

  while delivering substantial improvement, >11%, compared to the RAG baseline

**Efficiency in terms of tokens consumed**




**Efficiency in terms of API calls**



# Analysis

- **Effectiveness of adaptive generation**
  - The number of generated passages can be controlled by $\hat{m}$ , and results show that the system does not generate passages excessively

**Performance and averge number of generaed passages using different $\hat{m}$**

| | NQ | TriviaQA | BioASQ | PopQA | Overall | $m$ |
|---|---|---|---|---|---|---|
| $\hat{m}$=1 | 52.20 | 84.10 | 60.14 | 44.38 | 61.71 | 0.69 |
| $\hat{m}$=2 | 52.20 | 85.16 | 60.84 | 43.26 | 62.00 | 1.24 |

# Analysis

- **Impact of Source-Awareness**
  - Conduct an ablation study where source labels (own memory and external retrieval) were removed during the consolidation and answer generation process
  - Provide source information (ASTUTE RAG) leads to better performance overall compared to omitting it (ASTUTE RAG (No Source))
  - Awareness of information origin aids the consolidation process

**Ablation study on source-awareness**

| Model | NQ | TriviaQA | BioASQ | PopQA | Overall |
|---|---|---|---|---|---|
| ASTUTE RAG | **50.2** | 81.6 | **58.0** | **40.5** | **59.2** |
| ASTUTE RAG (No Source) | 48.1 | 82.3 | 57.7 | 39.9 | 58.6 |

# Analysis

## • Prompt for Intermediate Step Evaluation

**Prompt for Intermediate Step Evaluation**

**Task:** You are provided with the following:
1. A question.
2. The correct answer.
3. The input context.
4. The model's response, which contains:
- Consolidated context.
- Confidence scores for candidate answers.

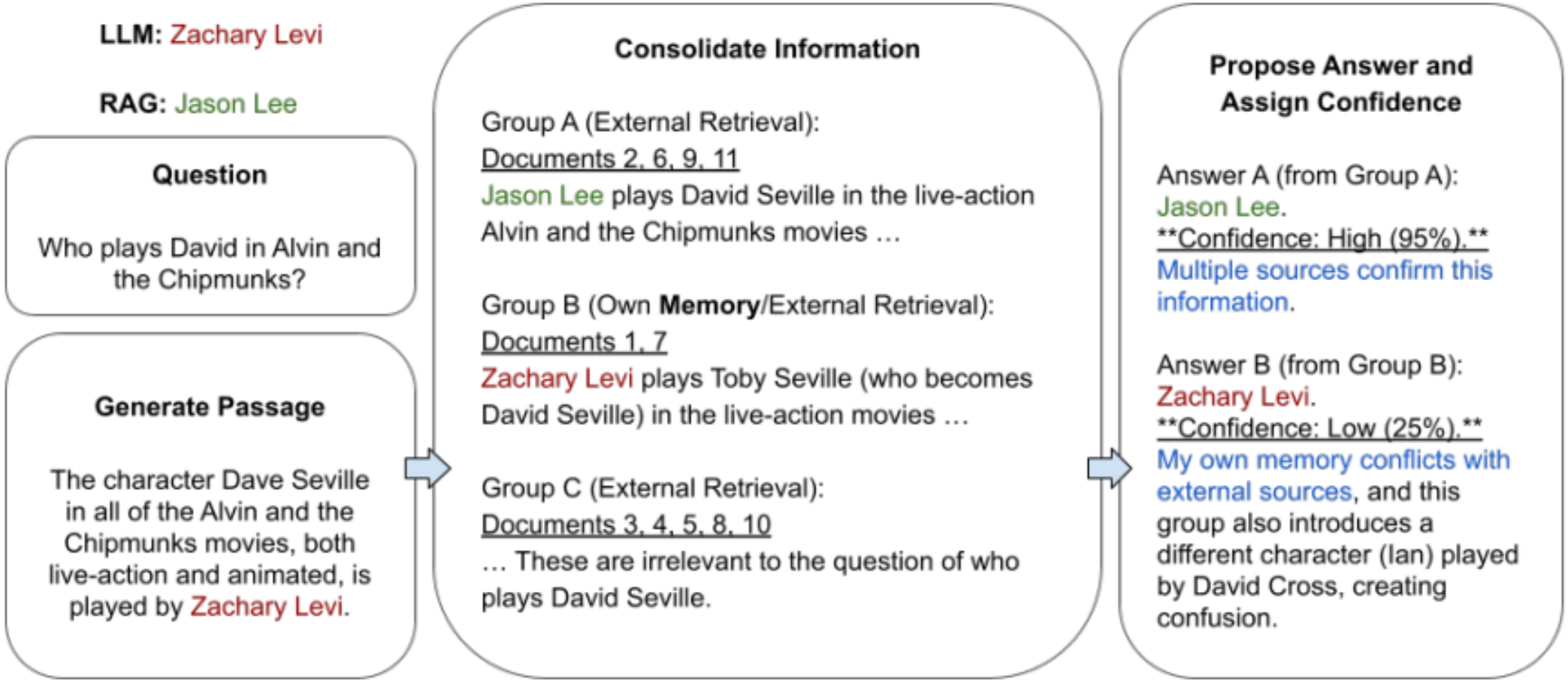
Your task is to:
- Evaluate the **quality of the consolidated context** in the model's response and provide a label: '<consolidation> correct </consolidation>' or '<consolidation> incorrect </consolidation>'. This evaluation is only about whether the consolidation is correct given the input context.
- Evaluate the **accuracy of the confidence score** (whether it aligns with the confidence of the supporting context) and provide a label: '<confidence> correct </confidence>' or '<confidence> incorrect </confidence>'. The evaluation is only based on the consolidated context.

Note that correct consolidation and confidence do not necessarily indicate the correct answer.
Question: {query}
Correct Answer: {answer}
Input Context: {input}
Model Response: {response}
Evaluation:

- To investigate the performance of intermediate steps, including knowledge consolidation and confidence assignment
- Use LLM-as-a-judge with the instruction in Appx. A.
- Our experimental results show that the accuracy for knowledge consolidation is 98.2%, and for confidence assignment, it is 95.0%.
- These results demonstrate the effectiveness of the proposed framework in the intermediate stages.

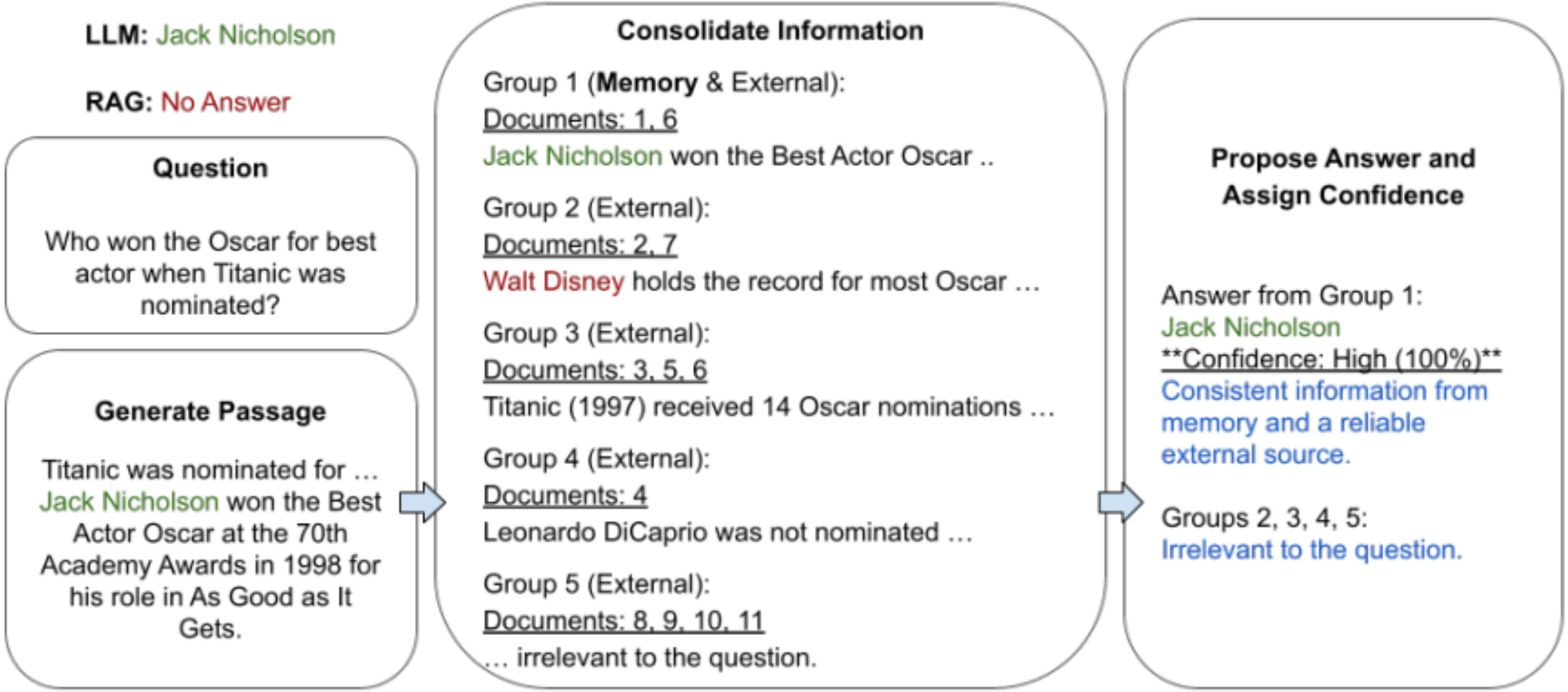
# Analysis

- **Qualitative examples**
  - Identiy the error in internal knowledge (i.e., generated passage) by confirming with external sources



# Analysis

- **Qualitative examples**
  - Detect the correct answer from imperfect retrieval by checking with its internal knowledge



# Conclusion

- **ASTUTE RAG**
  - Motive
    - Investigate the impact of imperfect retrieval on the performance of RAG systems
    - Identify knowledge conflicts as a key challenge
  - Contribution
    - Leverages the internal knowledge of LLMs and iteratively refines the generated responses
    - Consolidate internal and external knowledge in a source way
  - The effectiveness of ASTUTE RAG
    - Mitigate the negative effects of imperfect retrieval
    - Improve the robustness of RAG (with unreliable external sources)

# Conclusion

- **Limitations**
  - Effectiveness hinges on the capabilities of advanced LLMs with strong instruction following and reasoning abilities
  - Potentially more limited applicability with less sophisticated LLMs

- **Future Work**
  - Extending the experimental setup to include longer inputs would be important, where the challenges of imperfect retrieval and knowledge conflicts may be even more pronounced

# Appendix

- **Retrieval-augmented generation (RAG)**
  - RAG is commonly used for large language models (LLMs) to tackle **knowledge-intensive tasks** (Guu et al., 2020; Lewis et al., 2020)
  - **Knowledge-intensive Tasks (Prior works)**
    - Mainly leverage RAG to address the inherent knowledge limitations of LLMs
    - Effectively integrate missing information and grounding to reliable sources

**REALM (Guu et al., 2020)**

**Answer Likelihood**

$$p(y \mid x) = \sum_{z \in \mathcal{Z}} p(y \mid z, x)\, p(z \mid x)$$

**Knowledge Retriever**

$$p(z \mid x) = \frac{\exp f(x, z)}{\sum_{z'} \exp f(x, z')},$$

$$f(x, z) = \mathtt{Embed}_{\mathtt{input}}(x)^{\top} \mathtt{Embed}_{\mathtt{doc}}(z),$$

**Knowledge-Augmented Encoder**

$$p(y \mid z, x) \propto \sum_{s \in S(z,y)} \exp\left(\mathtt{MLP}\left(\left[h_{\mathtt{START(s)}}; h_{\mathtt{END(s)}}\right]\right)\right)$$

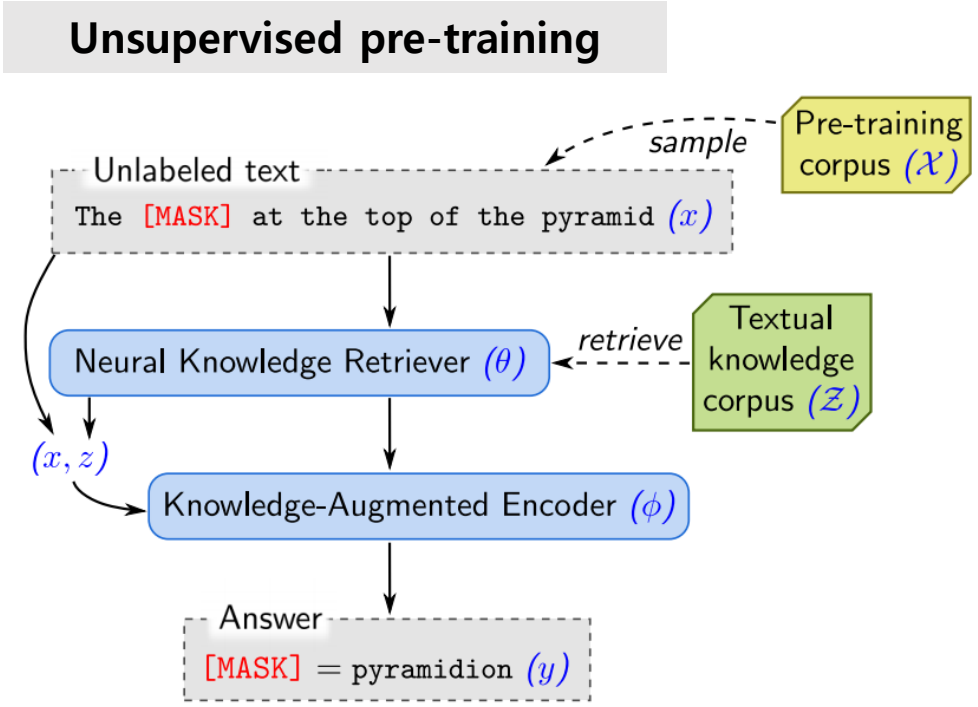
$$h_{\mathtt{START(s)}} = \mathtt{BERT}_{\mathtt{START(s)}}(\mathtt{join}_{\mathtt{BERT}}(x, z_{\mathrm{body}})),$$

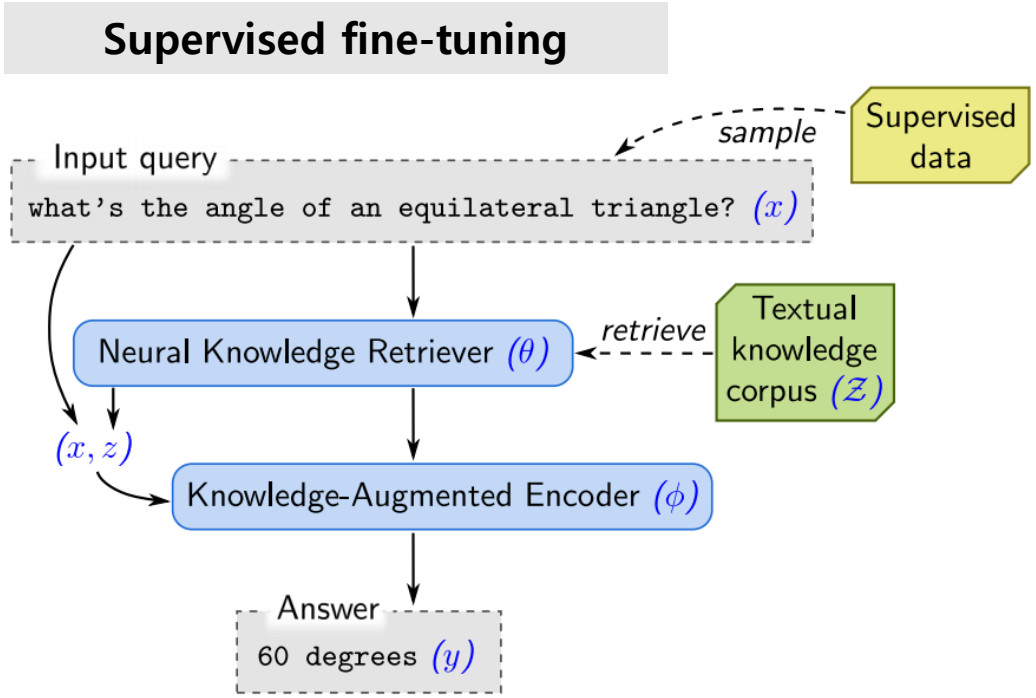
$$h_{\mathtt{END(s)}} = \mathtt{BERT}_{\mathtt{END(s)}}(\mathtt{join}_{\mathtt{BERT}}(x, z_{\mathrm{body}})),$$

Kelvin Guu et al. Retrieval augmented language model pre-training. ICML 2020.
Sunhao Dai et al. Unifying bias and unfairness in information retrieval: A survey of challenges and opportunities with large language models. WSDM 2025.

# Appendix

- **REALM (Guu et al., 2020)**
  - In contrast to models that store knowledge in their parameters, this approach explicitly exposes the role of world knowledge by asking the model to decide what knowledge to retrieve and use during inference




Kelvin Guu et al. Retrieval augmented language model pre-training. ICML 2020.
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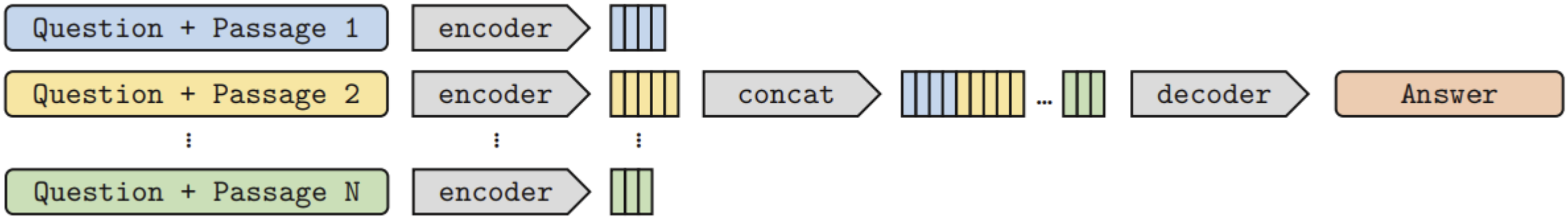
# Appendix

- **Fusion-in-Decoder (Izacard et al., 2021)**
  - Architecture

**Architecture of the Fusion-in-Decoder method**

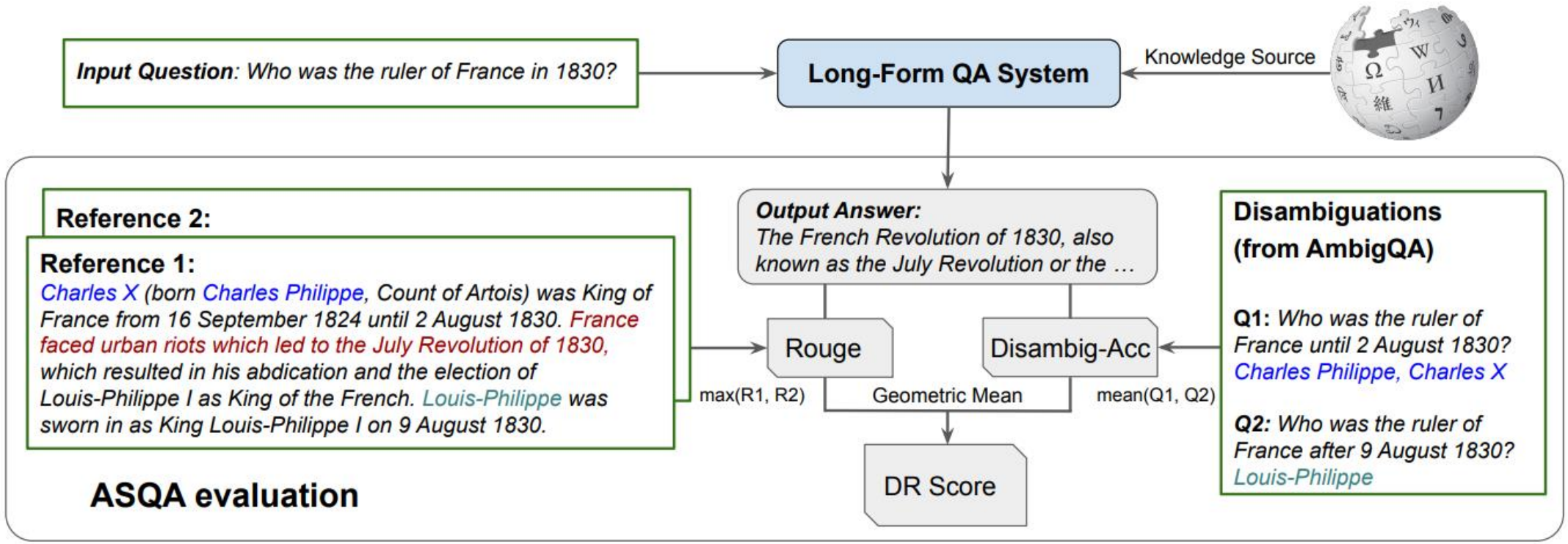





Gautier Izacard et al. Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering. EACL. 2021.

# Appendix

- **ASQA (Stelmakh et al., 2022)**
  - Long-form QA dataset for factoid questions
  - Focuses on answering ambiguous factoid questions






Ivan Stelmakh et al. Asqa: Factoid questions meet long-form answers. EMNLP. 2022.